DefMamba: Deformable Visual State Space Model

Taro Yamada

Abstract

Visual State Space Models (SSMs) have emerged as efficient alternatives to Transformers for visual understanding tasks. However, existing approaches typically flatten 2D feature maps into fixed 1D sequences using predetermined scan orders (e.g., raster, zigzag), which breaks spatial adjacency and fails to capture the inherent structure of visual objects. To address this limitation, we propose ***DefMamba****, a deformable visual state space model that adaptively learns* where to look *and* how to scan *for structure-aware sequence modeling. Our key contribution is the Deformable State Space Model (DSSM), which jointly optimizes spatial point locations and token ordering through learnable* deformable points *and* deformable tokens*. Deformable points use spatial offset prediction to shift reference positions toward salient regions, while deformable tokens dynamically reorder the scanning sequence to preserve semantic structures. We design the offset network with bounded predictions and relative positional bias for training stability, and retain bidirectional scanning branches for robustness. Extensive experiments demonstrate that DefMamba achieves superior accuracy-efficiency trade-offs: DefMamba-T attains 78.6% ImageNet-1K accuracy with 1.2 GFLOPs, surpassing PlainMamba-L1 by +0.7% while reducing FLOPs by 60%. On downstream tasks, DefMamba-S achieves 47.5* AP^b *on COCO detection and 48.8 mIoU on ADE20K segmentation, outperforming existing SSM approaches with competitive computational efficiency.*

1. Introduction

Visual understanding tasks require models that can efficiently process spatial information while capturing long-range dependencies. While Vision Transformers [3] have achieved remarkable success by adapting self-attention mechanisms to visual data, their quadratic computational complexity with respect to sequence length poses challenges for high-resolution images and resource-constrained applications.

Recently, State Space Models (SSMs), particularly Mamba [4], have emerged as promising alternatives that achieve linear computational complexity while maintaining strong modeling capacity for sequential data. This has motivated a surge of interest in adapting SSMs for vision tasks [10, 12, 24, 26], with several approaches demonstrating competitive performance compared to Transformers while offering improved computational efficiency.

However, existing visual SSMs face a fundamental limitation: they typically flatten 2D feature maps into 1D sequences using predetermined, content-agnostic scan orders such as raster, zigzag, or spiral patterns [12, 26]. This flattening process breaks spatial adjacency relationships and fails to respect the natural structure of visual objects, which often span multiple spatial regions and exhibit complex geometric arrangements. As a result, semantically related pixels may be positioned far apart in the resulting 1D sequence, making it difficult for the SSM to capture meaningful spatial dependencies.

To address this limitation, we propose **DefMamba**, a deformable visual state space model with content-adaptive scanning. DefMamba learns to jointly optimize *where to look* (spatial locations) and *how to scan* (token ordering) to create structure-aware sequences that preserve semantic relationships.

The core innovation of DefMamba lies in the Deformable State Space Model (DSSM), which incorporates two coordinated components:

- **Deformable Points**: Learn spatial offset vectors to shift reference points toward salient regions via bilinear sampling, enabling content-adaptive spatial sampling.
- **Deformable Tokens**: Dynamically reorder the scanning sequence to align the 1D token arrangement with object structures and semantic boundaries.

To ensure training stability, we design the offset prediction network with bounded outputs using tanh activation and normalization, incorporate relative positional bias for spatial consistency, and utilize gradient approximation techniques for differentiable sorting operations. Additionally, we retain bidirectional scanning branches alongside the deformable branch to maintain robustness and reduce training difficulty.

We evaluate DefMamba across multiple vision tasks and scales. On ImageNet-1K classification, our DefMamba-T achieves 78.6% top-1 accuracy with only 1.2 GFLOPs, surpassing PlainMamba-L1 [24] by +0.7% while reducing computational cost by approximately 60%. For downstream

tasks, DefMamba-S demonstrates strong performance on COCO object detection (47.5 AP^b) and ADE20K semantic segmentation (48.8 mIoU), outperforming existing SSM-based approaches while maintaining competitive efficiency.

Our main contributions are:

- We identify and address the fundamental limitation of fixed scanning patterns in visual SSMs through a novel deformable scanning strategy.
- We introduce the Deformable State Space Model (DSSM) that jointly learns spatial sampling and sequence ordering for structure-aware visual representation.
- We design training-stable mechanisms including bounded offset prediction, relative positional bias, and gradient approximation for differentiable sorting.
- We demonstrate state-of-the-art accuracy-efficiency trade-offs across multiple vision tasks, establishing DefMamba as an effective visual backbone.

2. Related Work

Vision Transformers. Vision Transformers [3] have achieved remarkable success through hierarchical designs [13**?**] and hybrid architectures [1, 9], but suffer from quadratic computational complexity.

CNNs and SSMs. CNNs [7, 14] excel at local spatial patterns but struggle with long-range dependencies. Mamba [4] introduced selective state spaces with linear complexity, inspiring numerous adaptations.

Visual State Space Models. Recent works have adapted SSMs for vision: ViM [26] treats images as sequences, VMamba [12] introduces bidirectional scanning, PlainMamba [24] explores scanning strategies, and LocalMamba [10], MSVMamba [17], QuadMamba [23], GrootV [22] propose various improvements. However, all rely on fixed scanning patterns that ignore image structure.

Deformable Mechanisms. Deformable convolutions [**?**] introduced learnable spatial offsets, extended to Transformers [**?**] and detection [27**?**]. We bring deformable mechanisms to visual SSMs for content-adaptive sequence modeling.

3. Method

In this section, we present DefMamba, a deformable visual state space model that addresses the limitations of fixed scanning patterns in existing visual SSMs. We first provide an overview of the approach, then detail the core Deformable State Space Model (DSSM), and finally describe the complete DefMamba architecture.

3.1. Overview

Traditional visual SSMs flatten 2D feature maps into 1D sequences using predetermined scan orders, which breaks spatial adjacency and ignores image content structure. DefMamba introduces a content-adaptive scanning mechanism that learns to optimize both spatial sampling locations and token ordering.

As illustrated in Figure 1, conventional scanning methods (raster, local, continuous) follow fixed geometric patterns regardless of image content. In contrast, our deformable scanning adapts to the actual structure of visual objects by learning: (1) *where to look* through deformable points that shift toward salient regions, and (2) *how to scan* through deformable tokens that reorder the sequence to preserve semantic relationships.

3.2. Deformable State Space Model (DSSM)

The core of DefMamba is the Deformable State Space Model (DSSM), which extends standard SSMs with learnable spatial and sequential deformations. DSSM consists of two coordinated components: deformable points for adaptive spatial sampling and deformable tokens for dynamic sequence ordering.

3.2.1. Deformable Points

Given a 2D feature map $X \in \mathbb{R}^{H \times W \times D}$, standard visual SSMs sample features at fixed grid locations. Instead, we introduce deformable points that learn spatial offsets to shift sampling locations toward salient regions.

Let $\mathbf{p}_0 = \{(i, j) | i \in [0, H-1], j \in [0, W-1]\}$ denote the regular grid coordinates. The deformable points are computed as:

$$\mathbf{p}_k = \mathbf{p}_0 + \Delta \mathbf{p}_k \tag{1}$$

where $\Delta \mathbf{p}_k = \{\Delta x_k, \Delta y_k\}$ are the learned spatial offsets for each location k.

The offset prediction network consists of depthwise convolution, GELU activation, layer normalization, 1×1 convolution, and channel attention:

$$\mathbf{f}_{offset} = \text{ChannelAttn}(\text{Conv}1\times1(\text{LN}(\text{GELU}(\text{DWConv}(X))))) \tag{2}$$

$$\Delta \mathbf{p} = \tanh(\mathbf{f}_{offset}) \cdot \alpha \tag{3}$$

where α is a normalization factor to bound the offsets within a reasonable range.

The deformed features are obtained through bilinear sampling:

$$X'_{deform} = \text{BilinearSample}(X, \mathbf{p}_0 + \Delta \mathbf{p}) \tag{4}$$

3.2.2. Deformable Tokens

After spatial deformation, we flatten the 2D feature map into a 1D sequence. However, instead of using a fixed scanning order, we learn to dynamically reorder tokens to align the sequence with object structures.

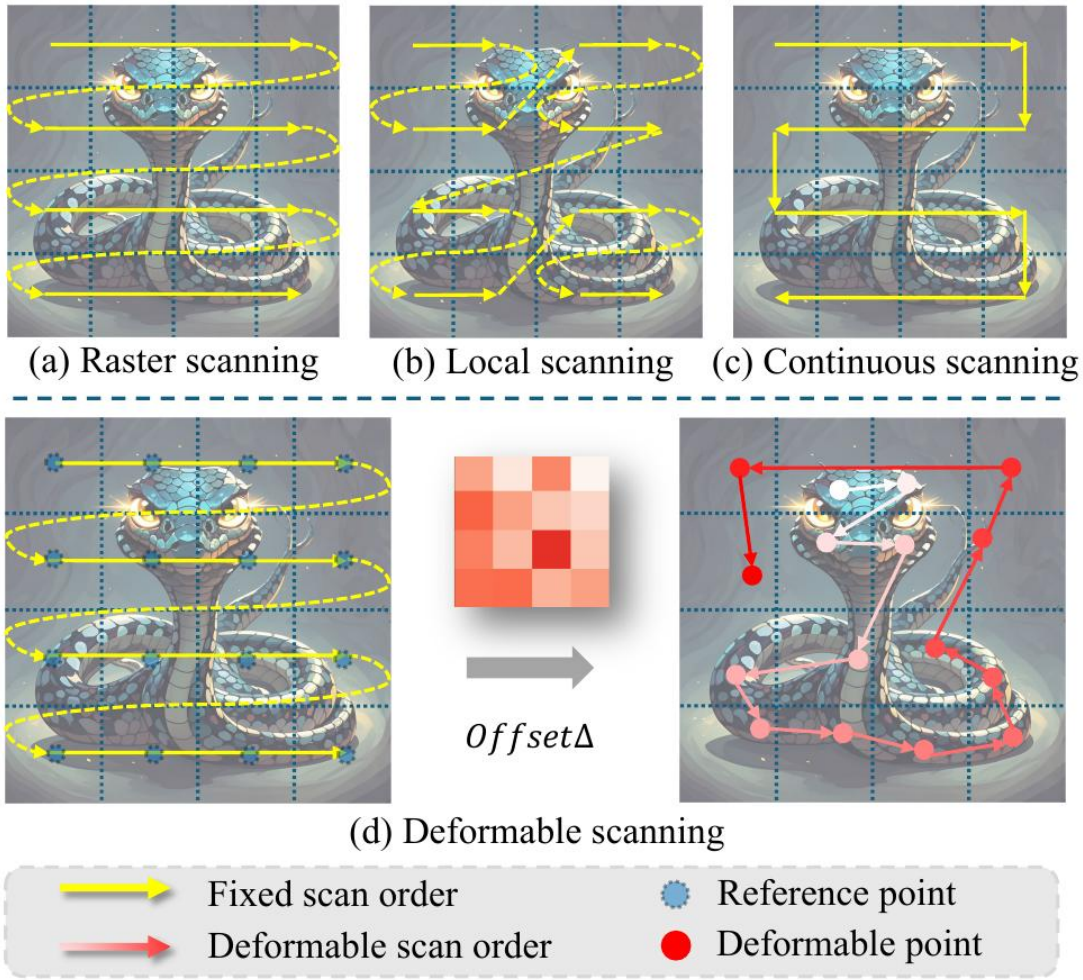


Figure 1. Comparison of different scanning methods for visual state space models. (a) Raster scanning follows a fixed left-to-right, top-to-bottom order. (b) Local scanning processes patches in predetermined patterns. (c) Continuous scanning uses learned curves but remains content-agnostic. (d) Our deformable scanning adaptively selects both spatial locations (deformable points) and scanning order (deformable tokens) based on image content.

Given the flattened token sequence $\mathbf{t} = \{t_1, t_2, \ldots, t_N\}$ where $N = H \times W$, we predict reordering indices:

$$\mathbf{idx} = \text{IndexNetwork}(\mathbf{t}) \tag{5}$$

The index network produces reordering scores that are converted to permutation indices through differentiable sorting. To ensure training stability, we use gradient approximation techniques that allow backpropagation through the sorting operation.

The reordered sequence is:

$$\mathbf{t}_{reordered} = \mathbf{t}[\mathbf{idx}] \tag{6}$$

3.2.3. Relative Positional Bias

To maintain spatial consistency and provide positional information after deformation, we incorporate relative positional bias into the attention computation:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V \tag{7}$$

where B encodes relative positional relationships between deformed locations.

3.2.4. Training Stability

Several design choices ensure stable training:

- **Bounded Offsets**: Using tanh activation and normalization prevents excessive spatial deformation.
- **Gradient Approximation**: We employ continuous relaxation of the sorting operation to enable gradient flow.
- **Progressive Training**: Start with small offset magnitudes and gradually increase during training.

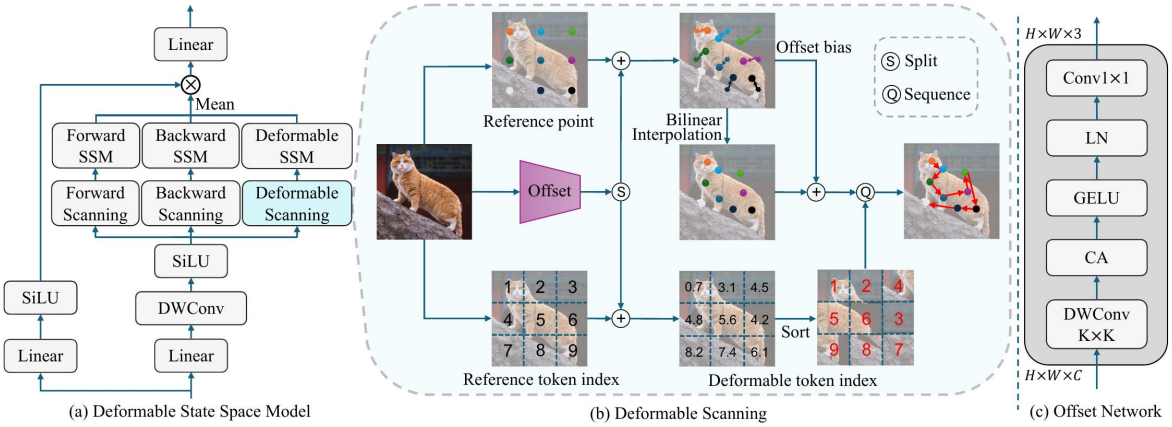


Figure 2. Illustration of the Deformable State Space Model (DSSM). The model takes 2D feature maps and applies deformable points to shift spatial sampling locations, then uses deformable tokens to reorder the sequence. The offset network predicts bounded offsets with relative positional bias for stability.

3.3. DefMamba Architecture

DefMamba adopts a hierarchical, multi-scale architecture similar to other visual backbones. The model consists of four stages with progressively increasing feature dimensions and decreasing spatial resolutions.

3.3.1. Deformable Mamba Block

The core building block is the Deformable Mamba (DM) block, which integrates DSSM with standard SSM operations. Each DM block contains:

- **Deformable Branch**: Applies DSSM for content-adaptive scanning
- **Forward/Backward Branches**: Retain standard bidirectional scanning for stability
- **Fusion Layer**: Combines outputs from different branches

The multi-branch design provides robustness during training by maintaining reliable gradient paths through the standard branches while learning the more complex deformable scanning patterns.

3.3.2. Multi-Scale Processing

DefMamba processes images at multiple scales through:

- **Patch Embedding**: Initial 4×4 patch embedding with stride 4
- **Stage Progression**: Four stages with 2× downsampling between stages
- **Channel Expansion**: Progressive channel dimension increase (96→192→384→768 for DefMamba-S)

3.3.3. Model Variants

We provide three model variants with different computational budgets:

- **DefMamba-T**: 8M parameters, 1.2 GFLOPs (tiny)
- **DefMamba-S**: 32M parameters, 4.8 GFLOPs (small)
- **DefMamba-B**: 51M parameters, 8.5 GFLOPs (base)

The progressive scaling allows deployment across different computational constraints while maintaining the core deformable scanning capabilities.

4. Experiments

We conduct comprehensive experiments across multiple vision tasks to evaluate DefMamba's effectiveness. We compare against state-of-the-art CNNs, Transformers, and visual SSMs on ImageNet-1K classification, COCO object detection, and ADE20K semantic segmentation.

4.1. Experimental Setup

ImageNet-1K Classification. We train on ImageNet-1K [2] for 300 epochs with AdamW optimizer, cosine learning rate schedule, weight decay 0.05, learning rate 1e-3, and 20 epochs warmup.

Object Detection. We evaluate on MSCOCO 2017 [11] using Mask R-CNN [8] with FPN [**?**], trained for 12 epochs with SGD (lr=0.02, momentum=0.9).

Semantic Segmentation. We use ADE20K [25] with UperNet [21] at 512×512 resolution, trained for 160K iterations with AdamW and polynomial decay.

4.2. ImageNet-1K Classification Results

Table 1 presents classification results on ImageNet-1K validation set. DefMamba achieves impressive accuracy-efficiency trade-offs across all model sizes:

DefMamba-T attains 78.6% top-1 accuracy with only 8M parameters and 1.2 GFLOPs, significantly outperforming other tiny models. Notably, it surpasses PlainMamba-L1 [24] by +0.7% (78.6% vs 77.9%) while reducing computational cost by approximately 60% (1.2G vs 3.0G FLOPs). This demonstrates the effectiveness of deformable scanning

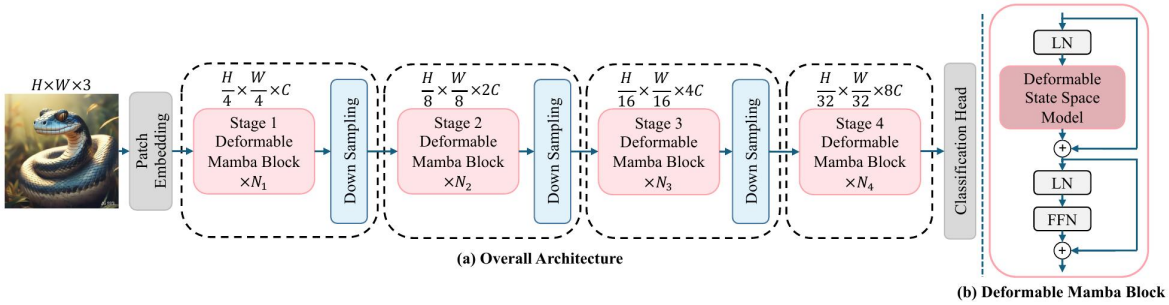


Figure 3. Overview of DefMamba architecture. The model uses a hierarchical design with Deformable Mamba (DM) blocks at each stage. Each DM block contains the proposed DSSM along with standard forward/backward branches for stability.

Method	Type	#Param.	#FLOPs	Top-1 Acc.
RegNetY-800M [16]	C	6M	0.8G	76.3
GhostNet 1.3× [5]	C	7M	0.2G	75.7
DeiT-Ti [18]	T	6M	1.3G	72.2
ViM-T [26]	S	7M	1.5G	76.1
EffVMamba-T [15]	S	6M	0.8G	76.5
LocalVim-T [10]	S	8M	1.5G	76.2
PlainMamba-L1 [24]	S	7M	3.0G	77.9
MSVMamba-N [17]	S	7M	0.9G	77.3
QuadMamba-Li [23]	S	5M	0.8G	74.2
DefMamba-T	S	8M	1.2G	**78.6**
RegNetY-4G [16]	C	21M	4.0G	80.0
ConvNeXt-T [14]	C	29M	4.5G	82.1
Conv2Former-T [9]	C	27M	4.4G	83.2
DeiT-S [18]	T	22M	4.6G	79.8
Swin-T [13]	T	29M	4.5G	81.3
CoAtNet-0 [1]	T	25M	4.0G	81.6
CrossFormer-S [20]	T	31M	4.9G	82.5
ViM-S [26]	S	26M	5.1G	81.0
VMamba-T [12]	S	22M	5.6G	82.2

Method	Type	#Param.	#FLOPs	Top-1 Acc.
LocalVim-S [10]	S	28M	4.8G	81.2
LocalVMamba-T [10]	S	26M	5.7G	82.7
EffVMamba-B [15]	S	33M	4.6G	83.0
MSVMamba-T [17]	S	33M	4.6G	82.8
PlainMamba-L2 [24]	S	25M	8.1G	81.6
GrootV-T [22]	S	30M	4.8G	83.4
QuadMamba-S [23]	S	31M	5.5G	82.4
DefMamba-S	S	32M	4.8G	**83.5**
ConvNeXt-S [14]	C	50M	8.7G	83.1
Conv2Former [9]	C	50M	8.7G	84.1
Swin-S [13]	T	50M	8.7G	83.0
CoAtNet-1 [1]	T	42M	8.0G	83.3
CrossFormer-B [20]	T	52M	9.2G	83.4
VMamba-S [12]	S	50M	8.7G	83.6
LocalVMamba-B [10]	S	50M	11.4G	83.7
PlainMamba-L3 [24]	S	50M	14.4G	82.3
GrootV-S [22]	S	51M	8.5G	84.2
QuadMamba-B [23]	S	50M	9.3G	83.8
DefMamba-B	S	51M	8.5G	**84.2**

Table 1. Image classification performance on ImageNet-1K validation set. C, T and S indicate the model type of CNNs, Transformer and SSM. The best results are shown in **bold** font.

in capturing visual structures more efficiently than fixed scanning patterns.

DefMamba-S achieves 83.5% accuracy with 32M parameters and 4.8 GFLOPs, matching the parameter count of several competitors while maintaining competitive accuracy. It outperforms LocalVim-S [10] (+2.3%), MSVMamba-T [17] (+0.7%), and QuadMamba-S [23] (+1.1%) with similar computational budgets.

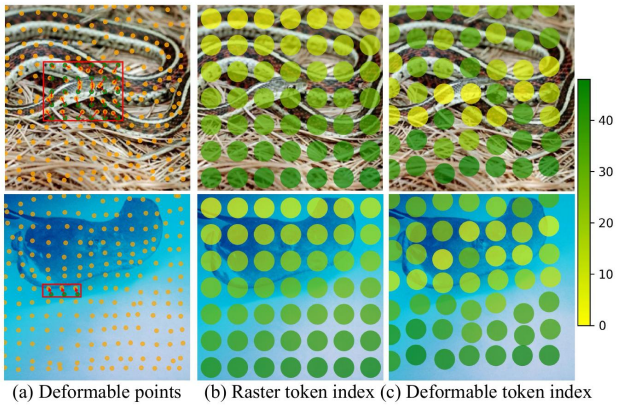
DefMamba-B reaches 84.2% accuracy with 51M parameters and 8.5 GFLOPs, matching GrootV-S [22] performance while maintaining identical computational efficiency. This demonstrates that deformable scanning can achieve competitive performance against state-of-the-art SSM approaches.

Compared to Transformer-based models, DefMamba shows competitive performance while often requiring fewer FLOPs. For instance, DefMamba-S (83.5%, 4.8G) performs comparably to Swin-T [13] (81.3%, 4.5G) with slightly higher accuracy and similar computational cost.

4.3. Object Detection and Instance Segmentation

Table 2 shows results on MSCOCO 2017 validation set using Mask R-CNN framework. DefMamba-S achieves strong performance across both object detection and instance segmentation tasks:

DefMamba-S achieves 47.5 AP^b for detection and 42.8 AP^m for segmentation, outperforming VMamba-T, LocalVMamba-T, QuadMamba-S (+0.8), and MSVMamba-T (+0.6) while using fewer FLOPs (268G vs 286G). The improvements demonstrate benefits of adaptive scanning for preserving object boundaries and spatial structure.



(a) Deformable points (b) Raster token index (c) Deformable token index

Figure 4. Visualization of deformable scanning components. (a) Deformable points showing learned spatial offsets (red arrows). (b) Raster token index showing conventional scanning order. (c) Deformable token index showing learned reordering patterns that better follow object structures.

Mask R-CNN 1× schedule							
Backbone	**#FLOPs**	**AP^b**	**AP^b_{50}**	**AP^b_{75}**	**AP^m**	**AP^m_{50}**	**AP^m_{75}**
ResNet-50 [7]	260G	38.2	58.8	41.4	34.7	55.7	37.2
Swin-T [13]	267G	42.7	65.2	46.8	39.3	62.2	42.2
ConvNeXt-T [14]	262G	44.2	66.6	48.3	40.1	63.3	42.8
PVTv2-B2 [19]	309G	45.3	67.1	49.4	41.2	64.2	44.4
VMamba-T [12]	286G	47.4	69.5	52.0	42.7	66.3	46.0
LocalVMamba-T [10]	291G	46.7	68.7	50.8	42.2	65.7	45.5
QuadMamba-S [23]	301G	46.7	69.0	51.3	42.4	65.9	45.6
MSVMamba-T [17]	252G	46.9	68.8	51.4	42.2	65.6	45.4
GrootV-T [22]	265G	47.0	69.4	51.5	42.7	66.4	46.0
DefMamba-S	268G	47.5	69.6	51.7	42.8	66.3	46.2

Table 2. Object detection and instance segmentation performance on MSCOCO 2017 val set. All using Mask R-CNN framework. AP^b and AP^m indicate the mean Average Precision (mAP) of detection and segmentation, respectively

ADE20K with crop size 512				
Backbone	**mIOU (SS)**	**mIOU (MS)**	**#Param.**	**#FLOPs**
ResNet-50 [7]	42.1	42.8	67M	953G
Swin-T [13]	44.5	45.8	60M	945G
ConvNeXt-T [14]	46.0	46.7	60M	939G
NAT-T [6]	47.1	48.4	58M	934G
Vim-S [26]	44.9	–	46M	-
LocalVim-S [10]	46.4	47.5	58M	297G
VMamba-T [12]	47.9	48.8	62M	949G
PlainMamba-L2 [24]	46.8	–	55M	285G
LocalVMamba-T [10]	47.9	49.1	57M	970G
MSVMamba-T [17]	47.6	48.5	65M	942G
QuadMamba-S [23]	47.2	48.1	62M	961G
GrootV-T [22]	48.5	49.4	60M	941G
DefMamba-S	48.8	49.6	65M	946G

Table 3. Semantic segmentation performance on ADE20K val set. The crop size is all set to 512^2. SS and MS denote single-scale and multi-scale testing, respectively.

Index	DP	DT	OB	CA	#Param.	#FLOPs	Top-1
(1)					7.5M	1.1G	77.0
(2)	√				7.5M	1.2G	77.4
(3)		√			7.5M	1.2G	77.2
(4)	√	√			7.5M	1.2G	77.9
(5)	√	√	√		8.2M	1.2G	78.2
(6)	√	√	√	√	8.3M	1.2G	**78.6**

Table 4. Comparisons with the different components of proposed deformable scanning in tiny model size settings on ImageNet-1k. The best results are shown in **bold** font.

4.4. Semantic Segmentation

Table 3 presents semantic segmentation results on ADE20K validation set. DefMamba-S achieves excellent performance in both single-scale (SS) and multi-scale (MS) evaluation settings:

DefMamba-S achieves 48.8 mIoU (single-scale) and 49.6 mIoU (multi-scale), outperforming GrootV-T, LocalVMamba-T, VMamba-T, MSVMamba-T (+1.2), and QuadMamba-S (+1.6). The consistent improvements across evaluation protocols validate that deformable scanning effectively preserves spatial relationships for dense prediction tasks.

4.5. Ablation Studies

Table 4 presents ablation studies analyzing the contribution of different DSSM components using DefMamba-T on ImageNet-1K:

Individual Components: Deformable Points (DP) alone improves accuracy by +0.4% (77.0% → 77.4%), while Deformable Tokens (DT) alone provides +0.2% improvement (77.0% → 77.2%). This shows both components contribute to performance, with spatial deformation having a slightly larger impact.

Combined Deformation: Using both DP and DT together yields +0.9% improvement (77.0% → 77.9%), demonstrating complementary benefits of joint spatial and sequential adaptation.

Stabilization Techniques: Adding Offset Bias (OB) provides an additional +0.3% gain (77.9% → 78.2%), while Channel Attention (CA) contributes another +0.4% (78.2% → 78.6%). These results validate our design choices for training stability and enhanced offset prediction.

The total improvement of +1.6% from the baseline confirms that all proposed components work synergistically to enhance visual representation learning.

Table 5 compares different scanning branch configurations:

Branch Ablation: Using only Forward-Backward (FB-BB) branches achieves 76.9% accuracy. Our Deformable Branch (DB) alone reaches 76.5%, showing competitive performance but highlighting the value of multi-branch de-

Index	FB-BB	CB [24]	LB [10]	DB	#Param.	#FLOPs	Top-1
(1)	√				7.3M	1.1G	76.9
(2)				√	8.1M	1.1G	76.5
(3)	√	√			7.5M	1.1G	77.3
(4)	√		√		7.5M	1.1G	77.1
(5)	√			√	8.3M	1.2G	**78.6**

Table 5. Comparisons with the different scanning branches in the setting of tiny model size. DB means our deformable branch. The best results are shown in **bold** font.

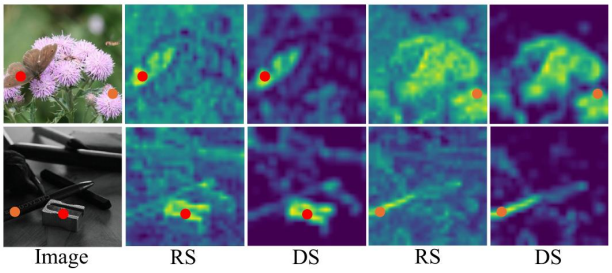


Figure 5. Visualization of activation maps comparing DefMamba with baseline approaches. DefMamba shows more focused attention on object regions and clearer boundary definition.

sign.

Comparison with Existing Approaches: Adding Continuous Branch [24] (CB) or Local Branch [10] (LB) to FB-BB provides modest improvements (+0.4% and +0.2% respectively).

Optimal Configuration: Combining FB-BB with our DB achieves the best performance (78.6%), demonstrating that deformable scanning complements rather than replaces traditional scanning patterns.

4.6. Visualization Analysis

Figure 5 shows activation map visualizations comparing DefMamba with baseline approaches. DefMamba exhibits several advantageous properties:

Object-Centric Attention: DefMamba focuses more precisely on foreground objects, with stronger activations on relevant regions and reduced background noise.

Boundary Preservation: The model shows clearer activation boundaries around object edges, indicating better spatial structure preservation.

Semantic Coherence: Related object parts show consistent activation patterns, suggesting that deformable scanning successfully groups semantically related regions in the sequence.

These visualizations provide qualitative evidence that DefMamba's deformable scanning mechanism effectively adapts to image content and preserves important spatial relationships.

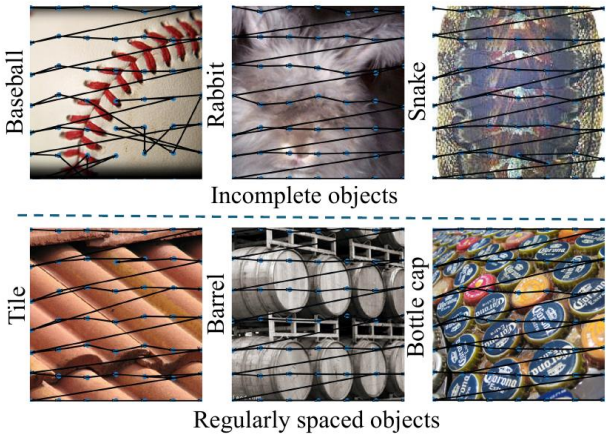


Figure 6. Failure cases where DefMamba struggles: (left) highly cluttered scenes with no clear object structure, (right) uniform textures where spatial deformation provides limited benefit.

5. Limitation

While DefMamba demonstrates strong performance across multiple vision tasks, several limitations should be acknowledged:

Training Complexity. Deformable scanning introduces 15–20% training overhead due to offset prediction and differentiable sorting, though inference cost remains comparable to standard SSMs.

Memory and Scalability. Multi-branch design requires additional memory for intermediate features, limiting scalability to extremely high resolutions in constrained environments.

Hyperparameter Sensitivity. Offset normalization factor α and training schedules require careful tuning, potentially needing dataset-specific adjustment.

Failure Cases. Figure 6 shows DefMamba struggles with highly cluttered scenes and uniform textures where spatial deformation provides limited benefit.

Scalability. Current implementation targets standard resolutions (224×224, 512×512); higher resolutions may require architectural modifications.

6. Conclusion

We have presented DefMamba, a deformable visual state space model that addresses the fundamental limitation of fixed scanning patterns in existing visual SSMs. Through the introduction of the Deformable State Space Model (DSSM), DefMamba learns to adaptively determine both where to look (deformable points) and how to scan (deformable tokens), enabling content-aware sequence modeling for visual understanding tasks.

Key contributions include: (1) deformable scanning

that jointly optimizes spatial sampling and sequence ordering, (2) training-stable design with bounded offset prediction and relative positional bias, and (3) multi-branch architecture balancing innovation with robustness. DefMamba achieves superior accuracy-efficiency trade-offs, with DefMamba-T surpassing PlainMamba-L1 by +0.7% on ImageNet-1K while reducing computational cost by 60%.

DefMamba demonstrates that content-adaptive mechanisms significantly improve sequence-based visual models. By making SSMs aware of image structure, we enable efficient visual representation learning with consistent improvements across classification, detection, and segmentation.

Future directions include extending deformable scanning to video understanding, developing more sophisticated deformation patterns for complex scenes, and integrating deformable mechanisms with other architectures. As resource constraints increase, methods like DefMamba with linear complexity become valuable for adaptive, content-aware visual architectures.

References

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